Universal Design in the Signage of Japan: Case Study of Visual Communication on the Subway

As matters of accessibility and equity become increasingly visible in contexts across the world, designers must understand how to create content which audiences of all backgrounds can interpret. Particularly, the ability to effectively communicate with multilingual communities in an increasingly globalized world is a necessity of inclusive design. Public signage in Japan presents a compelling example of universal design in practice, where designers must serve both a monoethnic population of residents and millions of foreign, non-Japanese-reading visitors.

The ubiquitous subway system of Japan, servicing 15 billion passenger journeys every year, requires passengers to rapidly interpret information about mobility and transportation (Schimkowsky, 2022). One manner of communication is signage, which can be posted anywhere with available wall surface on trains and in stations, acting as “constellations” of information (Cockain, 2018). In fact, these signs are so prevalent in Japanese metro systems that a genre of “manner posters” exists, defined by posters that encourage public behavior by use of “cultural references, elaborate drawings or cute characters” (Schimkowsky, 2023). Given the importance of passenger safety and efficiency in this massive transportation system, effective design is essential (Denis and Pontille, 2010).

The CRAP design principles, defined by Williams as contrast, repetition, alignment, and proximity, exist to identify elements of aesthetically pleasing designs (posters, infographics, websites, etc.) and will be my primary method of judging the effectiveness of signage in this paper (2014). Furthermore, I will identify possible strategies to effectively communicate information to passengers which may be adopted by designers serving the transportation systems of the United States.

Sample

The figures selected for analysis have been chosen because they successfully adhere to CRAP principles, almost exclusively use Japanese language, and are able to communicate their intended messages solely through images. Because the purpose of this case study is to solely analyze the ability of effective design, I have selected signage that is indecipherable to non-Japanese viewers. In other words, this is an analysis on visual literacy, not textual literacy.

The figures depicted in this analysis are photographs taken by me on my smartphone while using the subway system of Tokyo and Osaka. These figures all relate in some way to product advertisement, public etiquette, or travel information.

Procedure

Each sign will be accompanied by a description of its geographical placement to provide context and assessed based on its adherence to CRAP design principles. I also analyze the success of signage at communicating to an non-Japanese viewer under a section titled Universality, which I define as the quality of a design to be interpretable by a diverse linguistic audience. In the Discussion, I will identify common visual strategies which may be adopted by American designers.

Analysis & Results

Case Study 1: Poster of Future Transit

Figure 1: Poster depicting a development plan for the Osaka Metro Group.

This poster was found in the Shin-Osaka train station and features various urban locations and transportation devices with descriptive captions, all framed by a prominent date of 2035. From these details, this poster conclusively advertises future plans for the integration of advanced technology in the Osaka metro system.

Contrast

A viewer's eye is instantly drawn to the bright, large text "2035," communicating this is an image of future transit. The date sharply contrasts the dark blue background, standing out as the central focal point. Furthermore, the complexity of information increases as the viewer's eye wanders downward, weaving through skies with holograms and drones into streets laced with sleek buses and floor-to-ceiling windowed office buildings before arriving at subterranean train stations. This increasing complexity of information invites a viewer to spend more time with each layer, unraveling all that is planned for the Osaka Metro Group.

However, because of the sheer number of captions and scenes sharing the same space, fine details become lost. Subway stations jut into the roadway above ground, a human forearm overlays with a smartwatch overlays a train car, and it becomes a challenge to determine where information begins and ends.

Though visual clutter challenges a viewer's ability to interpret specific details in this metro scene, the poster as a whole is successful in communicating a wholistic, tech-prominent image of the future of public transportation in Osaka.

Repetition

The consistent use of captions and arrows to label information prevents the viewer from snagging on any unequal sections. An overarching blue gradient appears both in the background and in various objects, cleverly uniting all elements of the poster, regardless of their location or shape. Overall, these repeated elements make for a cohesive, futuristic theme which underlines this image of public transit in 2035.

Captions appear too frequently though, obscuring objects and inundating scenes with text that means nothing to a non-Japanese-reading viewer, risking frustration. Slim arrows slice through scenes that a viewer may want to examine without interference. Additionally, the use of a realistic, human forearm and smartwatch stands starkly opposed to the illustrative, colorful style used in the poster.

While this poster successfully incorporates futuristic elements across its design, it includes too many additional textual elements as well as an image of a smartwatch which unfortunately prevent a wholly cohesive piece.

Alignment

Major elements like the date 2035 and above ground, urban facilities appear centered, sprawling and utilizing the whole space the poster allows to portray a bustling, lively scene of

the future. This alignment creates a sense that no space is left unused, even blank space which splits the centered date in the sky from urban life on the ground. By replicating the physical layers of a city, this poster cleverly aligns information according to its geographic position in reality.

Some scenes in the bottom of the poster are angled at awkward positions, jostling for space and creating an unwelcome feeling of busyness. Particularly the lower right corner, containing a scene of people overseeing operations on wide screen panels, has a rounded border—the only one of its kind—which intrudes on a scene of pedestrians navigating the train station. As a result, the architecture of the design becomes somewhat dissonant, as scenes no longer neatly align with one another.

Were these scenes to neatly exist and naturally transitioning into each other, the alignment of this poster would not feel skewed towards the bottom of the design in contrast to the success of the scenes above ground.

Proximity

Related groups of information share proximity in this design as they would in reality, as this image neatly organizes scenes into levels in the sky, on ground, and below ground. As the eye trends lower, icons of public transportation share space with one another, from buses on roads to trains in stations. These distinct areas guide the viewer naturally through this poster's proposition of the future of public transportation.

However, the limited space of this design results in scenes intersecting one another, jostling for space. There is no white space in the lower portion of the design, only captions, pedestrians, technology, and sprawling metro stations.

Like with alignment, the overcrowding of information towards the bottom of the poster may cause a viewer to feel overwhelmed with so many scenes intersecting.

Universality

Overall, this poster excels in quickly communicating to a viewer that this design portrays a plan for the future of public transportation. Universal icons synonymous with modern technology such as pristine white and blue palettes, glass windows, and smart devices appear in every scene of this design. With the focal point being the date 2035, one can assume this poster means to reveal the future of public transportation in Osaka within seconds of engagement. Although elements remain busy, clashing for space at times, the depiction of an idealistic, tech-dominated future is apparent.

Case Study 2: Poster of Health and Travel Etiquette

Figure 2: Poster depicting proper behavior to prevent the spread of illnesses.

This poster was found in the Sumida City Skytree train station and features several images containing health and travel imagery arranged in an unordered sequence. These types of etiquette posters with a “detail-reduced” style are omnipresent in Japan, generalizing objects in order to draw attention to the depicted action rather than the objects themselves (Schimkowsky, 2022; Wilde, 2018). Through a minimalistic style and neatly arranged scenes, this poster educates viewers about proper etiquette concerning travel and illness.

Contrast

Despite its illustrative and muted style, this design effectively portrays six scenes that are visibly distinct from one another. In this way, key elements such as masks, handwashing, and symptoms of illness become the focal points of their scenes. For example, in the bottom left scene, a man who appears flustered faces away from his suitcase and bag, universal symbols of travel, which are overlaid by a glaring red X. The highlighting effect of red on both the man’s face and his travel items is apparent thanks to the pastel color scheme and simplistic style. The dark text appropriately stands out against the light blue background, also.

While this pastel color scheme is effective as a backdrop to highlighting desired behaviors on this poster, it reduces the poster's visual impact, making the design less eye-catching in a setting as bustling as a train station. In other words, it is unlikely a viewer would engage with this poster if it were at a fair distance from them.

Were a viewer to notice this unassuming, minimalistic poster, they would be drawn in by its effective use of contrast in the pairing of neat illustrations and dark text against a pleasing, pastel background.

Repetition

The design of this poster is incredibly cohesive, as all its illustrations possess the same line width, color scheme, and accompanying space for text. This consistent use of line drawings and muted colors ties all visual elements together, even though there is physical distance between each one. A clear hierarchy of information is maintained through these repeated design choices, allowing viewers to easily scan and interpret the poster.

The only elements were are not repeated are the topmost travel case icon and the bottommost attributions and logos. The travel case is effective in inviting white space on both its sides and further communicating this design's application to travel, though it is a bold visual element that does not reappear. The attributions and logos at the bottom, while perhaps necessary, are not rendered in the same blue color that all other text appears in, presenting a violation of consistency.

Overall, this poster contains more repeated elements than varying ones, though to a scrutinizing eye these differences may cause a viewer to become frustrated.

Alignment

The scenes are neatly arranged in a grid-like pattern, as each illustration and its corresponding text are aligned in two neat columns with an appropriate amount of white space to distinguish them. This alignment of content into columns and rows of two allows a viewer to easily and predictably navigate the design from the title, the first illustration on the left, then to the right, then to the next leftmost design in the row below. The result is a clean, organized, and accessible poster.

As mentioned in Repetition, the topmost travel case icon is somewhat disconnected from the content of the poster. The excess white space isolates it from the scenes below, and its position aligned with the center of the design juxtaposes it with the two-by-two pattern of its main body.

The order of events is also not immediately apparent, causing viewers to question if the scenes are supposed to happen in a particular order.

Regardless, because the scenes are mostly aligned into two neat columns with proper white space between all distinct items, viewers can easily scan and interpret the poster's content.

Proximity

Because the scenes and their text are grouped closely together, viewers quickly understand the correlation between them as cohesive items, especially with the ample white space that distances them from other scenes. The eye naturally identifies these distinct scenes and interprets them one at a time, reading across and down the page in order.

The content utilizes white space to organize the margins between each item so much that there is a slight sense of disconnection across the piece, like there is white space that could instead contain more information. Reducing the distance between sections could enhance the unity of the design, creating a more cohesive, integrated progression between scenes. It could cause the poster to have a more significant impact from a distance, too.

Universality

By using universal images like a suitcase, mask, bus, and calendar, non-Japanese-reading viewers can glean information about travel etiquette regarding illness without understanding the accompanying text below each image. Such universal practices as wearing a mask, maintaining social distance, and washing hands are recognized globally and easy communicated through the unassuming, minimalistic style of this poster. As a result, the clear and inviting presentation of information invites all viewers to engage with the design in a manner that global audiences can appreciate.

Discussion

This mini study asserts that universal design is essential in creating effective and inclusive communication tools, particularly in public spaces where diverse audiences interact. The public transit system of Japan serves as an excellent setting to analyze effective signage because of its ability to rapidly interact with and educate millions of people, native residents and foreign visitors alike. The two posters analyzed here, both photographed in major metropolitan train stations, demonstrate how universal design can transcend language, ensuring its messages are accessible to a wide demographic. In urban environments where people of varying ages, languages, and abilities converge, universal design is a necessity for safe, efficient, and equitable experiences.

Though each poster has its own deficiencies when assessed according to the CRAP principles, both rely heavily on iconography to communicate universal concepts. This sort of iconography depicts images of objects or behaviors that most individuals, regardless of background or literacy, can recognize. Suitcases equate to travel, masks communicate health, and urban buildings of glass appear futuristic. Red X's denote prohibited behavior and arrows indicate direction without the need for extensive text. By leveraging visual indicators and preexisting biases in viewers, these posters communicate complex behaviors and expectations in an easily digestible manner.

This analysis underscores relevance of universal design to urban scholars and designers of public signage. For urban planners, understanding how signage influences public behavior can inform the creation and jurisdiction of safer, more efficient public spaces in the same way that manner posters educate travelers in Japan. Similarly, designers working in public communication can draw inspiration from the effective use of both CRAP principles and universal iconography in these posters. Universal design is equitable; it improves everyone's lives, from the busy salaryman making his daily commute to the foreign backpacker riding a train for the first time.

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